



OPENER - Open-NER in domains without annotated resources

Emanuel Matos^{1,2,3}, Gabriel Silva^{1,2,3}, Mário Rodrigues^{1,3,4}, António Teixeira^{1,2,3}

¹IEETA, University of Aveiro, Aveiro, Portugal

²DETI, University of Aveiro, Aveiro, Portugal

³LASI – Intelligent System Associate Laboratory, Portugal

⁴ESTGA, University of Aveiro, Aveiro, Portugal

easm@ua.pt, grsilva@ua.pt, mjfr@ua.pt, ajst@ua.pt

Abstract

The development of Named Entity Recognition (NER) systems relies heavily on large annotated datasets, posing a significant challenge for new domains. This paper explores the effectiveness of LLM-based systems and automatically annotated data for NER development without manual annotations. Specifically, we propose using In-context Learning with a few examples included in prompts obtained from existing NER systems, which were trained without any manually annotated data. The proposed method was evaluated using two datasets (WikiNER and MiniHAREM), two LLMs (Bison and Sabiá-2), and two languages (Portuguese and Spanish). Median Precision scores exceeded 75%, with a maximum F1 score of 78.3% on the WikiNER PT dataset using Bison. In cross-domain experiments, with MiniHAREM, Precision remained high but Recall dropped to below 20%. Results also showed that more examples in prompts improve Precision; Bison outperforms Sabiá-2, particularly on Portuguese datasets; and both LLMs are capable of processing Spanish sentences having access to only examples in Portuguese. The results showed that automatic annotations with proper prompts and examples enable high-performance NER systems with cross-language and cross-domain potential. The proposed approach has high potential in terms of adaptability and speed of application to new domains.

Index Terms: Named Entity Recognition (NER), Automatic annotation, In-context learning, few-shot prompting, Large Language Models (LLM), cross-language adaptability.

1. Introduction

Named Entity Recognition (NER) [1, 2, 3] has significantly progressed with the introduction of deep learning (DL) and large language models (LLMs) [4]. In general, existing NER systems heavily depend on annotated data, posing several challenges, particularly in their development for new domains. The main issue is the reliance on extensive, domain-specific annotated datasets, which are expensive and labour-intensive to create. This reliance limits the application of NER in emerging fields where annotated data is limited or unavailable.

Recent evolution in Deep Learning and Large Language Models (LLMs) Deep learning techniques and large language models, such as BERT [5], GPT [6], and their variants, have revolutionised the field of natural language processing (NLP). These models have demonstrated remarkable capabilities in understanding and generating human language. The shift towards using LLMs for generative tasks has opened new possibilities for NER, by framing it as a text generation problem rather than a classification task [3]. This approach leverages the powerful contextual understanding of LLMs to improve the accuracy and flexibility of NER systems.

The effectiveness of NER models traditionally depends on the availability of large annotated datasets. Creating these datasets requires significant human effort and expertise, which translates to high costs and time consumption. This dependency is particularly problematic when expanding NER systems to new domains, where annotated data may not be available or may require extensive manual annotation efforts to create.

Besides the major problem of dependence of costly annotated data, other limitations must be addressed to increase availability of high quality NER systems for a richer set of languages and domains, being particularly relevant:

- NER models trained on one domain often perform poorly when applied to another. The specific terminology and context of new domains present challenges for generalisation.
- Identifying the exact boundaries of entities is complex, especially in languages with flexible word order or rich morphological structures, such as Portuguese.

Considering the recent evolutions in LLMs and their application to NLP tasks, particularly resorting to few-shot prompting [7] and In-context Learning (ICL) [8], the main objectives for this work were:

1. Explore and evaluate the effectiveness of using context and examples in prompts to perform NER based on LLMs.
2. Assess the performance that is attainable using only automatic annotations to obtain the content for the prompts.
3. Explore and compare different approaches to prompts, including ones based in markup languages such as XML [9].
4. Assess cross-language potential.

The paper presents 3 main contributions: The first, and more general, is the proposal of a novel NER method based on In-context learning. The proposed method, departing from common approaches, recognises an open set of tags, resulting in a novel form of NER, that we named Open-NER (OPENER for short) inspired by Open Information Extraction (Open-IE) [10]. It allows for a flexible and dynamic set of tags, enhancing potential to adapt to new domains. To the best of the authors' knowledge, this proposal is novel, not just for Iberian languages. A second contribution are the experimental results demonstrating that automatic annotations can be used to obtain the examples need for the prompts. This approach is cost-effective and scalable, facilitating the rapid development of NER systems for new domains. The different form of prompts, XML based, completes the set of contributions. It frames the NER problem as a generative task and uses XML annotations for examples used in prompts, exploiting the generative capabilities of LLMs.

2. Related work

Most relevant related work to contextualise the proposed method includes recent evolutions in NER, particularly for Portuguese, and In-context learning based NER.

2.1. Recent work in NER for Portuguese

Representative recent work in NER for Portuguese includes: frameworks for automatic NER creation [11]; BERT-based NER systems for Portuguese [12, 13, 14]; cross-domain explorations [15], proofs-of-concept using deep learning models [16], and fine-tuning BERT for new domains (e.g., legislative documents in Brazilian Portuguese [17]), all demonstrating significant improvements in Precision, Recall and potential applicability in real problems.

Particularly relevant for this paper is the research on NER development without the use of manually annotated data. A BERT-based NER for Portuguese trained with the results of automatic annotation by an ensemble of existing NER systems [11] demonstrated promising results in achieving F1 scores above 50% for domain-relevant classes such as LOCAL when trained only with automatically annotated data [12]. Matos et al. [16] expanded this methodology, demonstrating its applicability in different domains. By performing cross-domain evaluations and fine-tuning with within-domain data, the authors demonstrated that models initially trained on tourism data can effectively generalise to new domains, such as legal documents.

2.2. NER with In-Context Learning

Conventional NER methods rely on large domain-specific datasets, limiting their practicality. Recent approaches to NER use large language models (LLMs) and prompt-based heuristics for efficient, versatile NER without the need of large domain-specific datasets. Some representative examples of this line of research are briefly presented next.

GPT-NER [5], in 2019, transformed NER into a generation task, aligning it with LLMs capabilities, and exploring prompt designs, cross-language diversity, and performance in various contexts.

In 2023, Ashok et al. [18] introduced PromptNER, a cross-domain, few-shot NER algorithm using LLMs and prompt-based heuristics. It improved F1 metrics on CoNLL, GENIA, and FewNERD datasets, setting new benchmarks with a conditional generation model and few-shot examples.

Berger et al. [19] evaluated In-Context Learning (ICL) for NER in 2024, focusing on performance with limited annotation budgets. They found small annotated sample sets matched larger datasets' results, with strategies like "Highest Average" yielding high performance and random selection proving cost-effective.

These studies demonstrate the effectiveness of prompt-based heuristics and strategic sample selection for NER with minimal data, suggesting future research to refine these strategies across domains.

3. The Proposal: Generative Open-NER

Our proposal uses large language models (LLMs) and prompt-based heuristics to overcome the limitations of conventional NER methods, which typically require large, domain-specific datasets. In-context learning enables models to perform NER across multiple domains with minimal data by transforming

NER into a generation task, aligning with LLMs capabilities.

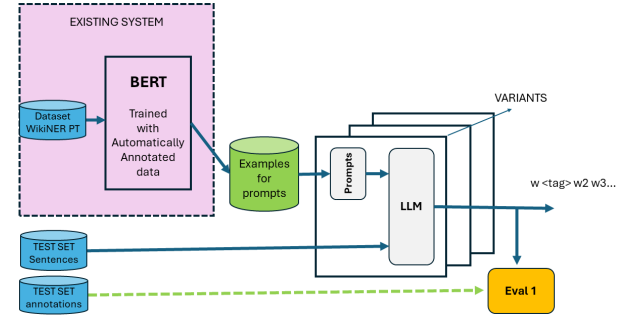


Figure 1: Overall architecture of the proposed method for Open-NER, showing the main blocks.

The Fig. 1 illustrates the proposed NER method. First, an existing NER automatic system uses BERT trained on automatically annotated data to generate a pool of examples. From this pool, examples are randomly chosen for few-shot learning in a prompt-based LLM system, with several variants resulting from different LLMs, prompts, and the number of examples used. The effectiveness of the method is evaluated by comparing the output of these LLMs against manual annotations in evaluation datasets. Additional relevant information follows regarding the first proof-of-concept of the method developed:

Large Language Models – As representative of the currently available models, two LLMs [20] were used: Google Bison [21] and Sabiá-2 (by Maritaca AI¹, a Brazilian company) [22].

Prompts – A variety of prompts was considered, resulting from 2 different approaches and number of examples.

The two approaches considered, resulting in what we designate as types, were: (Type 1) presenting the task as completion and providing examples of the entities followed by the tag; (Type 2) seeing the task as a generation task and providing as example the complete sentence with the chunks corresponding to entities delimited by XML opening and closing tags.

Regarding the number of examples included in prompts, three cases were considered: one example per class, two examples per class, and only one, from a randomly chosen class, that we call “1-shot”.

All the prompts had the same overall “classic” structure, starting with introduction, followed by examples and ending with the question [23, sec. 5.3]. Edited examples are presented in Table 1.

Examples for prompts – The set of examples for prompts were obtained by automatic annotation of a subset of WikiNER PT. The BERT-based annotation used by [12, 16] was selected for the automatic annotation. From the obtained annotations, the top 10 classes were extracted and used as tags.

Datasets – For a first instantiation of the proposed method the Portuguese part of the WikiNER dataset [24] was adopted as main dataset due to the availability of both manual annotations and an system for obtaining automatic annotation [16]. The dataset contains 7,200 manually labelled Wikipedia articles in nine languages, including Portuguese and Spanish. Additionally, to introduce domain and language diversity, the MiniHAREM and Spanish part of WikiNER were used. MiniHAREM dataset, resulting from the HAREM NER evaluation

¹<https://www.maritaca.ai/>

Table 1: Prompt structure and examples for the 2 types considered.

Part	Content example (Type 1)	Content example (Type 2)
Introduction: Translation:	Determine o tipo de entidade: <i>Determine the entity type:</i>	
Example: Translation:	Nova Iorque: CIDADE <i>New York: CITY</i>	Nas ruas de <CIDADE> Nova Iorque </CIDADE>. <i>In the streets of <CITY> New York </CITY>.</i>
Example: Translation:	Alexandre III: SER_HUMANO Rússia: PAÍS <i>Alexandre III: HUMAN BEING Russia: COUNTRY</i>	Outros atentados, como contra <SER_HUMANO> Alexandre III </SER_HUMANO> da <PAÍS> Rússia </PAÍS> ... <i>Other attacks, such as against <HUMAN_BEING> Alexandre III</HUMAN_BEING> from <COUNTRY> Russia</COUNTRY> ...</i>
Question: Translation:	As primeiras são comuns na parte equatorial da América do Sul ou em áreas litorâneas <i>The first ones are common in the equatorial part of South America or in coastal areas.</i>	

initiative for Portuguese language [25, 26], provides 21,901 manually tagged words.

Considering the size of the smaller corpus, MiniHAREM, to ensure equal conditions, only 2,000 test sentences were used from the each of the 3 datasets. They were selected by simple random sampling [27].

All examples used in the prompts were obtained from automatic annotation of the Portuguese part of the WikiNER dataset (excluding the 2,000 sentences randomly selected for testing), using 10 classes to ensure a comprehensive evaluation.

4. Experimental Evaluation

Aligned with the objectives, a set of variants of the proposed method were considered and compared to identify the most effective approaches for NER in different contexts. The exploration included investigating the impact of LLM adopted, type of prompt and number of examples included as well the effect of dataset, and language. Additionally, to assess repeatability of results, for one of the LLM, the experiments were repeated. The combination of these several factors resulted in a total of 45 evaluations.

The models' performance was assessed using Precision, Recall, and F1 metrics [1, 28], commonly adopted in the area. As the set of entities resulting by the proposed method is different from the manual annotations used as ground truth, the 3 metrics were calculated for each token - in general corresponding to a word - based only on the presence or absence of BIO tags. To allow these calculations, results obtained were post-processed to transform them to one token per line with the associated BIO tag. This simplification was considered adequate for an initial exploration of the potential of the proposed method.

5. Results

This section summarizes the results starting by analysis of the general performance results and continuing with analysis of the effect on the results of factors such as type of prompt.

5.1. Overall Performance

To assess the general level of performance attained by the proposed method, overall results for the 3 metrics on the 2 datasets are presented in Fig. 2.

The figure shows: medians for Precision above 75%; major impact of dataset in Recall and as consequence in F1, with median results for WikiNER above 50% and much worst for MiniHAREM. The mean values for Spanish are consistently lower than those for Portuguese, but the differences are not substantial.

The maximum F1 score obtained of 78.3% was achieved with Bison, 1-Shot and Type1 prompts on the WikiNER PT dataset, pointing to a good ability to correctly identify and clas-

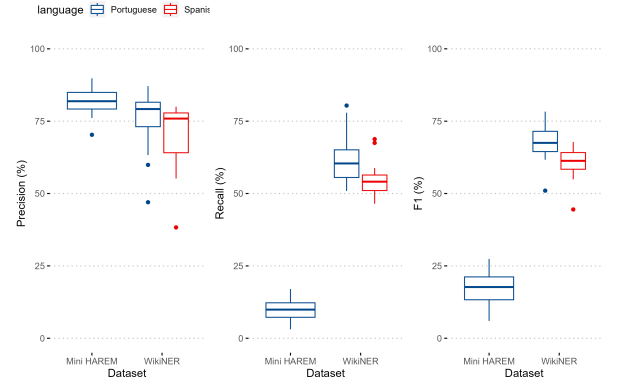


Figure 2: Overall performance metrics (Precision, Recall and F1) obtained on the 2 datasets and languages considered in experimental evaluation.

sify named entities when given a single example. On the MiniHAREM (PT) dataset, the performance was much lower, with the max F1 score being 22.2% using 1-Shot, suggesting difficulties in generalisation to this dataset.

The second LLM, Sabiá-2, best performance, with an maximum F1 score of 66.5% was obtained with 1 example per class on the WikiNER PT dataset, corroborating LLMs effectiveness with minimal training examples.

5.2. Effect of number of shots

The effect of the number of examples included in prompts in Precision and Recall is presented in Fig. 3. Results are presented separately by dataset (MiniHAREM and WikiNER) and language (SP and PT). It is clear from that using only one example in prompts is insufficient for optimal performance, as indicated by the lower Precision scores of 1-Shot. In general, the 2 variants that incorporate more examples exhibit higher Precision scores, with improvements observed in both languages. Also noticeable is the low Recall, below 20%, for MiniHAREM dataset.

5.3. Effect of Prompt Type

The analysis of type of prompt effect, in Fig. 4, indicates that Type 1 prompts yield higher recall than XML-based prompts (Type 2), across both datasets and languages. For Precision, results are more similar. Median values on MiniHAREM are similar and slightly higher for Type 2, but having Type 2 a much higher variance. In summary, Type 2 obtains higher maximum and median values in Precision while Type 1 is better in terms of Recall.

In both Precision and Recall, the results obtained on Spanish dataset remains consistently lower compared to Portuguese dataset.

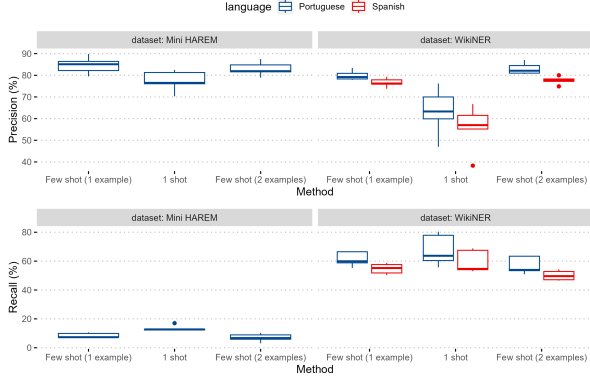


Figure 3: *Effect of examples in prompts in Precision (top) and Recall (bottom), separately for each dataset and language.*

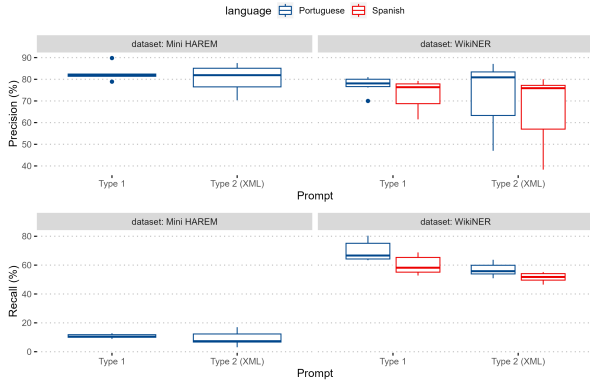


Figure 4: *Effect of the type of prompts in Precision (top) and Recall (bottom), separately for each dataset and language.*

5.4. Effect of the Large Language Model

The performance comparison of two large language models (LLMs), Bison and Sabiá-2, on the two datasets and languages, is presented in Fig. 5.

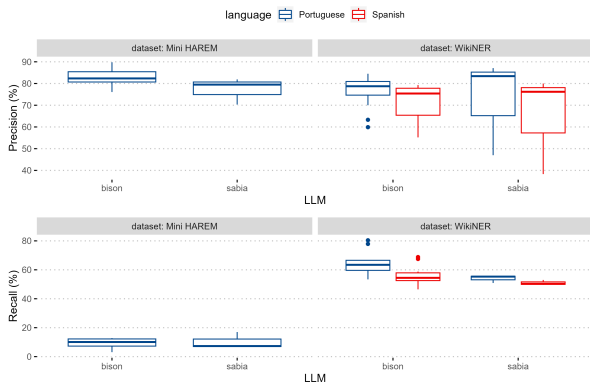


Figure 5: *Effect of number of the LLM in Precision (top) and Recall (bottom), separately for each dataset and language.*

The Bison model generally outperforms the Sabiá-2 model in terms of both Precision and Recall across both datasets and languages. An exception is the best maximum and median value for Precision on WikiNER being achieved by Sabiá-2, but with

higher variance of results. The performance gap between Bison and Sabiá-2 is more pronounced in the MiniHAREM dataset, particularly for Precision.

Both models perform better on the Portuguese datasets compared to the Spanish datasets. There is a noticeable drop in performance when the models are applied to the Spanish datasets, especially in terms of recall.

5.5. Cross-language potential

As the examples used were always from WikiNER PT, the experiments with Spanish assess cross-language potential. For the Bison model, maximum scores of 67.8% and 79.3% were obtained on WikiNER for F1 and Precision, respectively. The Sabiá-2 model attained maximums of 61.3% in F1 and 80.0% in Precision. Results in both models point to their capability to effectively transfer knowledge across languages.

5.6. Repetition and Consistency

The results in Fig. 6 indicate that precision and recall metrics do not vary much with repetition, both for Portuguese and Spanish.

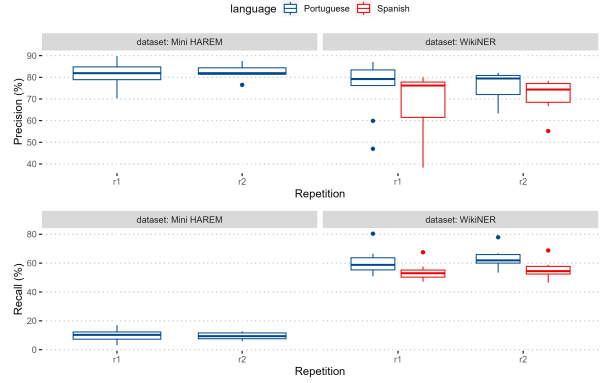


Figure 6: *Effect of repetition in Precision and Recall.*

6. Conclusion

This paper explores LLMs and automatically annotated data for NER development without manual annotations, proposing the use of prompts integrating a few examples obtained from existing NER systems, trained without any manually annotated data.

The results demonstrated the effectiveness of the proposed method. Key findings include the superior performance of Type 1 prompts over XML, the general superiority of the Bison model compared to the Sabiá-2 model, the interesting capability of models to generalise across languages and, to lower extent, domains (Portuguese to Spanish and WikiNER PT to MiniHAREM datasets), the improvement of performance with inclusion of more examples. Not needing manual annotated data, the proposed method enables simpler development of NER for new domains at reduced costs.

Considering the most promising results and identified limitations, future work should include: (1) Refining prompt strategies to optimise performance, starting by exploring the combination of the 2 types of prompts; (2) Integration of additional contextual information in prompts; (3) Development of automatic annotation systems similar to the one developed for WikiNER PT for the other datasets, to enable use of in-domain examples; (4) Exploring new datasets, domains and languages.

7. Acknowledgments

This work was partially supported by FCT (Foundation for Science and Technology) project reference 10.54499/UIDB/00127/2020 and DOI identifier <https://doi.org/10.54499/UIDB/00127/2020>. Gabriel Silva was supported by National Funds through the FCT (Foundation for Science and Technology) UI/BD/153571/2022 PhD scholarship. Access to Google’s LLMs was provided and by the RCNA project FCT/CPCA/2021/01.

The authors would like to thank ISCA and the organising committees of past Interspeech conferences for kindly providing the $\LaTeX$ template and acknowledge the use of ChatGPT to improve the clarity and coherence of the sentences in this document.

8. References

- [1] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed., 2024, ch. 8, Sequence Labeling for Parts of Speech and Named Entities.
- [2] P. Sun, X. Yang, X. Zhao, and Z. Wang, “An overview of named entity recognition,” in *2018 International Conference on Asian Language Processing (IALP)*, 2018, pp. 273–278.
- [3] A. Goyal, V. Gupta, and M. Kumar, “Recent named entity recognition and classification techniques: a systematic review,” *Computer Science Review*, vol. 29, pp. 21–43, 2018.
- [4] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, and J. Gao, “Large language models: A survey,” 2024. [Online]. Available: <https://arxiv.org/abs/2402.06196>
- [5] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” 2019. [Online]. Available: <https://arxiv.org/abs/1810.04805>
- [6] G. Yenduri, M. Ramalingam, G. C. Selvi, Y. Supriya, G. Srivastava, P. K. R. Maddikunta, G. D. Raj, R. H. Jhaveri, B. Prabadevi, W. Wang, A. V. Vasilakos, and T. R. Gadekallu, “GPT (generative pre-trained transformer)— a comprehensive review on enabling technologies, potential applications, emerging challenges, and future directions,” *IEEE Access*, vol. 12, pp. 54 608–54 649, 2024.
- [7] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei, “Language models are few-shot learners,” in *Proceedings of the 34th International Conference on Neural Information Processing Systems*, ser. NIPS ’20. Red Hook, NY, USA: Curran Associates Inc., 2020.
- [8] X. Liu, Y. Zheng, Z. Du, M. Ding, Y. Qian, Z. Yang, and J. Tang, “GPT understands, too,” *AI Open*, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666651023000141>
- [9] R. Eito-Brun, *XML-based Content Management: Integration, Methodologies and Tools*. Chandos Publishing, 2017.
- [10] G. Stanovsky, I. Dagan, and Mausam, “Open IE as an intermediate structure for semantic tasks,” in *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, C. Zong and M. Strube, Eds. Beijing, China: Association for Computational Linguistics, Jul. 2015, pp. 303–308. [Online]. Available: <https://aclanthology.org/P15-2050>
- [11] E. Matos, M. Rodrigues, P. Miguel, and A. Teixeira, “Towards automatic creation of annotations to foster development of named entity recognizers,” in *10th Symposium on Languages, Applications and Technologies (SLATE 2021)*. Schloss Dagstuhl-Leibniz-Zentrum für Informatik, 2021.
- [12] E. Matos, M. Rodrigues, and A. Teixeira, “Assessing Transfer Learning and automatically annotated data in the development of Named Entity Recognizers for new domains,” in *Proc. IberSPEECH 2022*, 2022, pp. 191–195.
- [13] F. Souza, R. Nogueira, and R. Lotufo, “BERTimbau: pretrained BERT models for Brazilian Portuguese,” in *9th Brazilian Conference on Intelligent Systems, BRACIS, Rio Grande do Sul, Brazil, October 20-23, 2020*.
- [14] —, “Portuguese Named Entity Recognition using BERT-CRF,” 2020, arXiv:1909.10649.
- [15] E. Matos, M. Rodrigues, and A. Teixeira, “Named entity extractors for new domains by transfer learning with automatically annotated data,” in *International Conference on Computational Processing of the Portuguese Language*. Springer, 2022, pp. 288–298.
- [16] —, “Towards the automatic creation of NER systems for new domains,” in *Proceedings of the 16th International Conference on Computational Processing of Portuguese*, 2024, pp. 218–227.
- [17] R. O. Nunes, D. G. Balreira, A. S. Spritzer, and C. M. D. S. Freitas, “A Named Entity Recognition Approach for Portuguese Legislative Texts Using Self-Learning,” in *Proceedings of the 16th International Conference on Computational Processing of Portuguese*, 2024, pp. 290–300.
- [18] D. Ashok and Z. C. Lipton, “PromptNER: Prompting for named entity recognition,” *arXiv preprint arXiv:2305.15444*, 2023.
- [19] U. Berger, T. Baumel, and G. Stanovsky, “In-context learning on a budget: A case study in named entity recognition,” 2024. [Online]. Available: <https://arxiv.org/abs/2406.13274>
- [20] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong *et al.*, “A survey of large language models,” *arXiv preprint arXiv:2303.18223*, 2023.
- [21] Z. B. Akhtar, “From Bard to Gemini: An investigative exploration journey through Google’s evolution in conversational AI and generative AI,” *Computing and Artificial Intelligence*, vol. 2, no. 1, pp. 1378–1378, 2024.
- [22] T. S. Almeida, H. Abonizio, R. Nogueira, and R. Pires, “Sabiá-2: A New Generation of Portuguese Large Language Models,” 2024. [Online]. Available: <https://arxiv.org/abs/2403.09887>
- [23] J. Berryman and A. Ziegler, *Prompt Engineering for LLMs: The Art and Science of Building Large Language Model-Based Applications*. O’Reilly Media, Inc., 2024.
- [24] J. Nothman, N. Ringland, W. Radford, T. Murphy, and J. R. Curran, “Learning multilingual Named Entity Recognition from Wikipedia,” *Artificial Intelligence*, vol. 194, pp. 151–175, 2012. [Online]. Available: <http://dx.doi.org/10.1016/j.artint.2012.03.006>
- [25] D. Santos, N. Seco, N. Cardoso, and R. Vilela, “HAREM: An advanced NER evaluation contest for Portuguese,” in *Proceedings of the 5th International Conference on Language Resources and Evaluation (LREC)*, N. Calzolari, K. Choukri, A. Gangemi, B. Maegaard, J. Mariani, J. Odijk, and D. Tapias, Eds., 2006.
- [26] C. Mota and D. Santos, Eds., *Desafios na avaliação conjunta do reconhecimento de entidades mencionadas: O Segundo HAREM*. Linguatca, 2008, ISBN: 978-989-20-1656-6. [Online]. Available: <http://www.linguatca.pt/LivroSegundoHAREM>
- [27] F. Olken and D. Rotem, “Random sampling from databases: a survey,” *Statistics and Computing*, vol. 5, pp. 25–42, 1995.
- [28] R. Yacoub and D. Axman, “Probabilistic Extension of Precision, Recall, and F1 Score for More Thorough Evaluation of Classification Models,” in *Proceedings of the First Workshop on Evaluation and Comparison of NLP Systems*, S. Eger, Y. Gao, M. Peyrard, W. Zhao, and E. Hovy, Eds. Online: Association for Computational Linguistics, Nov. 2020, pp. 79–91. [Online]. Available: <https://aclanthology.org/2020.eval4nlp-1.9>